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https://github.com/jesus-garcia-izquierdo/diatomic-chain

Diatomic chain: acoustic and optical branches, phonon density of states and heat capacity

Numerical heat capacity versus the Debye and Einstein models, displacement fluctuations and validation by Langevin dynamics

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Technical and theoretical documentation (English version)

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Abstract

This document collects the theory and the results of the `diatomic-chain` repository. The presentation goes *from the most general to the most particular*: we start from the description of a solid as a mechanical problem (the five elements: object[E1], fields[E2], coupling[E3], stage[E4] and observer[E5]), arrive at the harmonic crystal and its statistical-mechanical description in the canonical ensemble as an *ideal* system of oscillators (phonons), introduce the density of states (DOS) and the Einstein and Debye models, and only then specialise to the one-dimensional diatomic chain with two masses m_1, m_2 and, in general, two spring constants K_1, K_2. For it we obtain in closed form the acoustic and optical branches, the DOS and the cumulative number of modes; we compute numerically (Gauss–Legendre quadrature over the Brillouin zone) the heat capacity, internal energy, entropy and free energy; and we compare them quantitatively with the Debye model, the Einstein model and a hybrid Debye + Einstein model. We also discuss other observables: the effective Debye temperature $\Theta_D(T)$, the fluctuations of the atomic displacements (absence of long-range positional order in 1D) and bond fluctuations. Everything is validated along an independent route: a Langevin dynamics simulation (BAOAB integrator) from which we extract the DOS (velocity spectrum), the heat capacity (fluctuation–response relation) and the displacement correlations.

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Notation and units

Unless stated otherwise we work in **reduced units** $\hbar = k_{\mathrm{B}} = 1$, so an angular frequency, an energy and a temperature are measured in the same unit; the natural scale is $\sqrt{K/m_1}$. To convert to physical units recall that typical phonon energies are $\hbar\omega \sim 10$–$100\,\mathrm{meV}$, i.e. characteristic temperatures $\Theta = \hbar\omega/k_{\mathrm{B}} \sim 10^2$–$10^3\,\mathrm{K}$.

Symbol	Meaning
$m_1,\ m_2$	masses of the two atoms of the basis (sublattices $s = 1, 2$)
$K_1,\ K_2$	spring constants of the intra-cell and inter-cell bonds
a	lattice constant (length of the two-atom unit cell)
N	number of cells; $2N$ atoms and $2N$ normal modes
$\sigma = K_1 + K_2$, $\mu = m_1 m_2/(m_1 + m_2)$, $M = m_1 + m_2$	sum of spring constants, reduced mass, cell mass
$u_{s,n}$	displacement of atom s of cell n from its equilibrium position
k	wave number in the first Brillouin zone $(-\pi/a, \pi/a]$
$\omega_-(k),\ \omega_+(k)$	acoustic and optical branches
v_s	sound velocity, $v_s = a\sqrt{K_1 K_2/[(K_1 + K_2)M]}$
$g(\omega)$	phonon DOS per cell, $\int_0^\infty g\,\mathrm{d}\omega = 2$
$N(\omega)$	cumulative number of modes per cell, $N(\omega) = \int_0^\omega g$
$\beta = 1/k_{\mathrm{B}}T$	inverse thermal energy of the bath
C_V	heat capacity at fixed external parameters (here, fixed length)
$\Theta_D,\ \Theta_E$	Debye and Einstein temperatures

Part I. General framework

1 What we describe and with which ingredients

1.1 State, observables and dynamics

Every mechanical theory answers three questions: *what* information is enough to specify the system now (**state**), *which* quantities can be measured (**observables**) and *how* the state changes in time (**dynamics**). A crystalline solid has about 10^{23} degrees of freedom, so the complete microscopic description (a point of classical phase space, or a quantum state vector) is unmanageable and irrelevant for macroscopic purposes. Statistical mechanics replaces the microstate by a *macrostate* $\mathcal{B}$ characterised by a probability distribution over microstates (the ensemble), and the macroscopic values of the observables are averages over it. This project is a particularly clean example of that programme, because in the harmonic approximation the microscopic dynamics is solved *exactly* and the statistics reduce to those of an ideal system.

1.2 The five elements applied to a solid

Before writing any equation it pays to separate the five elements of the physical problem: an object[E1] (what moves), some fields[E2] (what acts on it), a coupling[E3] (how they act), a stage[E4] (the background structure, with its symmetries) and an observer[E5] (the choice of variables and approximations used to read the problem). Much conceptual confusion comes from mixing them.

Element	In a crystal (and in the diatomic chain)
object[E1]	The nuclei (with their core electrons) of masses m_s; state variables: displacements u and momenta p.
fields[E2]	The effective potential $V(\{\boldsymbol{R}\})$ created for the nuclei by the valence electrons in the Born–Oppenheimer approximation.
coupling[E3]	The forces $-\partial V/\partial \boldsymbol{R}$; in the harmonic approximation, linear springs K_1, K_2 between nearest neighbours.
stage[E4]	The Bravais lattice with a two-atom basis: *discrete translation* symmetry by multiples of a.
observer[E5]	Chosen representation: atomic displacements, normal coordinates, Bloch waves $\lvert k\rangle$ or phonon occupation numbers.

The moral that guides the whole document is the chain

$$\text{stage}^{\text{E4}} \implies \text{symmetries} \implies \text{conserved quantities and good quantum numbers.}$$

In the crystal, the discrete translation symmetry of the stage[E4] turns the wave number k (the *crystal momentum*) into a good quantum number and lets us diagonalise the problem block by block (Bloch's theorem).

1.3 System and environment: the crystal in contact with a bath

A real crystal is never isolated: it exchanges energy with its environment. We model it as a *closed system* in contact with a much larger heat bath, with a weak coupling whose only role is to allow heat exchange. Once equilibrium is reached the macrostate is $\mathcal{B} = (\beta, N, L)$ and the distribution of microstates is the **canonical** one,

$$\rho_{\mathcal{B}} \propto \mathrm{e}^{-\beta \hat{H}}, \qquad Z(\beta) = \operatorname{tr} \mathrm{e}^{-\beta \hat{H}}, \qquad F = -\beta^{-1} \ln Z. \tag{1}$$

From Z follow the macroscopic values:

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}, \qquad C_V = k_{\mathrm{B}} \beta^2 \frac{\partial^2 \ln Z}{\partial \beta^2}, \qquad S = \frac{\langle E \rangle - F}{T}. \tag{2}$$

In the thermodynamic limit the microcanonical and canonical ensembles coincide and the relative energy fluctuation vanishes (§3.4).

2 From the real solid to the harmonic crystal

2.1 Equilibrium configuration and expansion of the potential

The first step is static: the equilibrium configuration $\{\boldsymbol{R}^0\}$ makes the forces vanish, $\partial V/\partial \boldsymbol{R} = 0$. Writing $\boldsymbol{R} = \boldsymbol{R}^0 + \boldsymbol{u}$ and expanding,

$$V = V_0 + \frac{1}{2} \sum_{i,j} \Phi_{ij}\, u_i u_j + \mathcal{O}(u^3), \qquad \Phi_{ij} = \left. \frac{\partial^2 V}{\partial u_i\, \partial u_j} \right|_0 . \tag{3}$$

The **force-constant matrix** Φ is symmetric and, if the equilibrium is stable, positive semidefinite. Invariance of the stage[E4] under rigid translations imposes the *acoustic sum rule* $\sum_j \Phi_{ij} = 0$: moving the whole crystal costs no energy.

2.2 Small oscillations and normal coordinates

With $T = \frac{1}{2}\sum_i m_i \dot{u}_i^2$ Newton's equations read $M\ddot{\boldsymbol{u}} = -\Phi \boldsymbol{u}$ with $M = \mathrm{diag}(m_i)$. Looking for solutions $\boldsymbol{u} \propto \boldsymbol{v}\, \mathrm{e}^{-\mathrm{i}\omega t}$ leads to the generalised eigenvalue problem

$$\boxed{\det\left(\Phi - \omega^2 M\right) = 0} \tag{4}$$

which is the same mathematics that reappears with quantum observables: diagonalising a Hermitian operator. It is convenient to symmetrise it with the **dynamical matrix** $D = M^{-1/2}\Phi M^{-1/2}$, Hermitian and positive semidefinite, whose eigenvalues are $\omega^2 \geq 0$ and whose orthonormal eigenvectors $\boldsymbol{e}_\lambda$ define the **normal coordinates** Q_λ. In them

$$H = \sum_\lambda \left(\frac{P_\lambda^2}{2} + \frac{\omega_\lambda^2 Q_\lambda^2}{2} \right), \tag{5}$$

i.e. the harmonic crystal is an **ideal system**: a collection of *decoupled*, *distinguishable* oscillators. All the physics of the interactions is hidden in the spectrum $\{\omega_\lambda\}$.

Key idea

Anharmonicity ($\mathcal{O}(u^3)$) is the *residual interaction* of this ideal system. Energetically it is small, but conceptually it is essential: without it each normal mode would keep its energy forever, the dynamics would be integrable and an isolated realisation of the crystal would *not* be ergodic nor mixing (the famous Fermi–Pasta–Ulam–Tsingou experiment shows that a weak anharmonicity is not always enough either). In the canonical ensemble this is no problem: the heat bath thermalises each mode separately. In the simulation of §10 the Langevin bath plays exactly that role.

2.3 Discrete translation symmetry: Bloch's theorem

For a crystal of N cells with p atoms per cell in d dimensions, D has dimension dpN. Discrete translation symmetry lets us look for Bloch-wave solutions,

$$u_{s,n}(t) = \frac{e_s}{\sqrt{m_s}}\, \mathrm{e}^{\mathrm{i}(kna - \omega t)}, \tag{6}$$

which reduce the problem to a $dp \times dp$ matrix $D(k)$ for each k in the first Brillouin zone. With Born–von Kármán periodic boundary conditions, $u_{s,n+N} = u_{s,n}$, the allowed wave numbers are

$$k_j = \frac{2\pi j}{Na}, \qquad j = -\tfrac{N}{2}+1, \dots, \tfrac{N}{2}, \tag{7}$$

exactly N values. For each k there are dp frequencies $\omega_b(k)$: the **branches** of the dispersion relation.

- d **acoustic** branches, with $\omega \to 0$ as $k \to 0$: they are the manifestation of the acoustic sum rule (Goldstone modes of the translation symmetry broken by the crystal). At long wavelength they are sound waves, $\omega \simeq v_s |k|$.
- $d(p-1)$ **optical** branches, with $\omega(0) \neq 0$: the atoms of the basis oscillate against each other. If they carry different charges, these modes couple to light (hence the name).

In total there are dpN modes, as many as degrees of freedom.

3 Statistical mechanics of the harmonic crystal

3.1 Quantisation: phonons

Quantising each normal oscillator,

$$\hat{H} = \sum_{k,b} \hbar\omega_b(k)\left(\hat{n}_{k,b} + \tfrac{1}{2}\right), \qquad \hat{n}_{k,b} = 0, 1, 2, \ldots \tag{8}$$

The excitation number $n_{k,b}$ is interpreted as the number of **phonons** in mode (k, b). Phonons are bosons whose number is *not conserved* (the bath can freely create and destroy them), so their chemical potential vanishes and their mean number follows Bose–Einstein statistics with $\mu = 0$:

$$\langle n \rangle = \frac{1}{\mathrm{e}^{\beta\hbar\omega} - 1}. \tag{9}$$

3.2 Canonical ensemble: everything factorises

Being an ideal system of distinguishable oscillators, the partition function factorises,

$$Z(\beta) = \prod_{k,b} \frac{1}{2\sinh(\beta\hbar\omega_b(k)/2)}, \qquad \ln Z = -\sum_{k,b} \ln\Big[2\sinh\Big(\tfrac{\beta\hbar\omega_b(k)}{2}\Big)\Big], \tag{10}$$

and any macroscopic quantity is a sum of single-mode contributions. With $y = \hbar\omega/k_{\mathrm{B}}T$, the single-oscillator functions are:

Quantity	Quantum	Classical ($y \to 0$)
mean energy	$e = \dfrac{\hbar\omega}{2}\coth\dfrac{y}{2} = \dfrac{\hbar\omega}{2} + \hbar\omega\,\langle n\rangle$	$k_{\mathrm{B}}T$
heat capacity	$c = k_{\mathrm{B}}\,\dfrac{(y/2)^2}{\sinh^2(y/2)}$	k_{B}
free energy	$f = k_{\mathrm{B}}T\ln[2\sinh(y/2)]$	$k_{\mathrm{B}}T\ln y$
entropy	$s/k_{\mathrm{B}} = y\,\langle n\rangle - \ln(1-\mathrm{e}^{-y})$	$1 - \ln y$

$c(y)$ is the *Einstein function*. Two limits summarise it. If $T \gg \Theta_{\mathrm{vib}} = \hbar\omega/k_{\mathrm{B}}$ the oscillator is classical and contributes k_{B} (equipartition theorem: $k_{\mathrm{B}}/2$ per quadratic term of the Hamiltonian). If $T \ll \Theta_{\mathrm{vib}}/10$ the oscillator is almost always in its ground state: it is **frozen** and does not contribute to the heat capacity. The whole phenomenology of the heat capacity of solids can be read as the progressive "unfreezing" of the modes as temperature rises.

Summing over the dpN modes, the classical limit gives the **Dulong–Petit law**: $C_V = dpNk_{\mathrm{B}}$, independent of masses and spring constants. In our chain ($d = 1$, $p = 2$): $C_V/Nk_{\mathrm{B}} \to 2$, i.e. k_{B} per atom.

3.3 Phonon density of states

When $N \to \infty$ the k values form a continuum. The continuum limit $\frac{1}{N}\sum_k \to \frac{a}{2\pi}\int_{\mathrm{BZ}} \mathrm{d}k$ turns sums over modes into integrals, and it is convenient to group modes by frequency through the **density of states** (DOS) per cell:

$$g(\omega) = \frac{1}{N}\sum_{k,b} \delta(\omega - \omega_b(k)) = \frac{a}{2\pi}\sum_b \int_{\mathrm{BZ}} \mathrm{d}k\;\delta(\omega - \omega_b(k)), \qquad \int_0^\infty g(\omega)\,\mathrm{d}\omega = dp. \tag{11}$$

With it, any additive quantity per cell is

$$\frac{X(T)}{N} = \int_0^\infty g(\omega)\,x(\omega, T)\,\mathrm{d}\omega, \tag{12}$$

with x any function of the table above. All the microscopic information needed by the thermodynamics of the harmonic crystal is contained in $g(\omega)$.

In one dimension, $\delta(\omega - \omega(k))$ with $\omega(k)$ monotonic on $[0, \pi/a]$ gives

$$g(\omega) = \frac{a}{\pi} \sum_b \left| \frac{\mathrm{d}\omega_b}{\mathrm{d}k} \right|^{-1}_{\omega_b(k)=\omega}, \tag{13}$$

which diverges wherever the group velocity vanishes (band edges): these are the **van Hove singularities**, of type $(\omega_c - \omega)^{-1/2}$ in 1D and integrable.

3.4 Fluctuations and thermodynamic limit

In the canonical ensemble the energy fluctuates, and its variance is tied to a response function (fluctuation–response relation):

$$(\Delta E)^2 = \frac{\partial^2 \ln Z}{\partial \beta^2} = k_\mathrm{B} T^2 C_V. \tag{14}$$

Since C_V is extensive, $\Delta E / \langle E \rangle \sim N^{-1/2} \to 0$: in the thermodynamic limit the energy is sharp and the canonical and microcanonical ensembles are equivalent. This same relation is used in the simulation to *measure* C_V.

3.5 Asymptotic behaviour

High temperature. Expanding $c(y) = 1 - \frac{y^2}{12} + \frac{y^4}{240} - \dots$,

$$\frac{C_V}{N k_\mathrm{B}} = dp \left[1 - \frac{\hbar^2 \langle \omega^2 \rangle}{12\,(k_\mathrm{B} T)^2} + \mathcal{O}(T^{-4}) \right], \qquad \langle \omega^2 \rangle = \frac{1}{dp} \int g\, \omega^2 \, \mathrm{d}\omega. \tag{15}$$

The approach to Dulong–Petit is controlled by the *second moment* of the DOS, which is exact and simple: $\langle \omega^2 \rangle = \frac{1}{dp} \operatorname{tr} D(k)$, independent of k.

Low temperature. Only modes with $\hbar\omega \lesssim k_\mathrm{B} T$ are unfrozen; these are the small-k acoustic modes, with $g(\omega) \propto \omega^{d-1}$. Hence $C_V \propto T^d$: Debye's T^3 law in 3D and, in our chain,

$$\frac{C_V}{N k_\mathrm{B}} \simeq \frac{a}{\pi v_s} \int_0^\infty c(y)\, \mathrm{d}\omega = \frac{\pi}{3} \frac{a\, k_\mathrm{B} T}{\hbar v_s}, \qquad T \to 0, \tag{16}$$

where we used $\int_0^\infty y^2 \mathrm{e}^y / (\mathrm{e}^y - 1)^2 \, \mathrm{d}y = \pi^2/3$. The result depends only on the sound velocity: at low temperature the crystal behaves as a continuous elastic medium.

4 Reference models

Before computers, computing $g(\omega)$ was out of reach and people resorted to models that replace the real DOS by a simple one. The two classical ones each capture one end of the problem.

4.1 Einstein model

Einstein assumes all atoms oscillate independently with the same frequency ω_E: dpN distinguishable normal modes with equal frequency,

$$g_E(\omega) = dp\, \delta(\omega - \omega_E), \qquad \frac{C_V}{N k_\mathrm{B}} = dp\, \frac{(\Theta_E/2T)^2}{\sinh^2(\Theta_E/2T)}, \qquad k_\mathrm{B} \Theta_E = \hbar \omega_E. \tag{17}$$

It recovers Dulong–Petit and explains why $C_V \to 0$ as T decreases (a historical result: the first success of quantisation outside black-body radiation), but it predicts an *exponential* decay $C_V \sim \mathrm{e}^{-\Theta_E/T}$ in conflict with the observed power law. The reason is clear from the DOS: the model has no low-frequency modes.

Choice of Θ_E. A principled choice, instead of fitting it, is to require the model to reproduce the exact correction (15): $\omega_E^2 = \langle\omega^2\rangle$.

4.2 Debye model

Debye treats the whole crystal as an elastic medium: linear dispersion $\omega = v_s|k|$ up to a cutoff ω_D chosen so that there are exactly dpN modes. The DOS is $g_D \propto \omega^{d-1}$ up to ω_D; in one dimension it is *flat*:

$$g_D(\omega) = \frac{dp}{\omega_D}\,\Theta(\omega_D - \omega), \qquad \frac{C_V}{Nk_\mathrm{B}} = dp\,\mathcal{D}_1\!\left(\frac{\Theta_D}{T}\right), \qquad \mathcal{D}_1(x_D) = \frac{1}{x_D}\int_0^{x_D} \frac{y^2\mathrm{e}^y}{(\mathrm{e}^y-1)^2}\,\mathrm{d}y. \tag{18}$$

Since its low-frequency DOS is the exact one, the Debye model is *exact* for $T \to 0$ provided the true v_s is used. In 1D with two modes per cell, $g_D(0) = a/(\pi v_s)$ requires

$$\omega_D = \frac{2\pi v_s}{a}. \tag{19}$$

Its weakness is at intermediate frequencies: it ignores van Hove singularities, the gap and the actual shape of the branches.

Effective Debye temperature. A standard way to quantify the failure is to define, at each T, the value $\Theta_D(T)$ that would make the model exact: $dp\,\mathcal{D}_1(\Theta_D(T)/T) = C_V^{\mathrm{exact}}(T)/Nk_\mathrm{B}$. If Debye were exact, $\Theta_D(T)$ would be constant. Its limits are known: $\Theta_D(0) = 2\pi\hbar v_s/(ak_\mathrm{B})$ and, comparing second moments ($\langle\omega^2\rangle_D = \omega_D^2/3$ in 1D),

$$\Theta_D(\infty) = \frac{\hbar}{k_\mathrm{B}}\sqrt{3\,\langle\omega^2\rangle}. \tag{20}$$

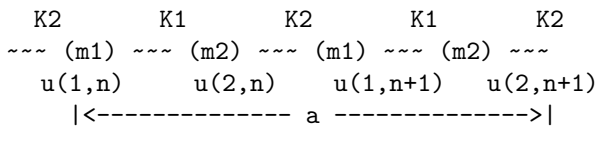
4.3 Hybrid Debye + Einstein model

For a crystal with a basis the natural idea is to separate branches: Debye for the d acoustic branches (with the exact v_s and a cutoff ω_D^{ac} holding dN modes) and Einstein for the $d(p-1)$ optical branches, with $\omega_E = \langle\omega_+\rangle_{\mathrm{BZ}}$. This is the model used in practice for solids with several atoms per cell, and we will see that its quality depends on how *flat* the optical branches are.

Part II. Specialisation: the diatomic chain

5 The model

We now set $d = 1$, $p = 2$. Cell n contains an atom of mass m_1 (displacement $u_{1,n}$) and one of mass m_2 ($u_{2,n}$) joined by a spring K_1; atom 2 of cell n is joined to atom 1 of cell $n+1$ by a spring K_2:



The textbook case is $K_1 = K_2 = K$; keeping $K_1 \neq K_2$ costs nothing and lets us discuss that the gap can be opened either by a mass contrast or by a stiffness contrast (dimerised chain). The Hamiltonian is

$$H = \sum_n \left[\frac{p_{1,n}^2}{2m_1} + \frac{p_{2,n}^2}{2m_2} + \frac{K_1}{2}(u_{2,n} - u_{1,n})^2 + \frac{K_2}{2}(u_{1,n+1} - u_{2,n})^2 \right], \tag{21}$$

and the equations of motion are

$$m_1 \ddot{u}_{1,n} = K_1(u_{2,n} - u_{1,n}) + K_2(u_{2,n-1} - u_{1,n}), \tag{22}$$
$$m_2 \ddot{u}_{2,n} = K_1(u_{1,n} - u_{2,n}) + K_2(u_{1,n+1} - u_{2,n}). \tag{23}$$

6 Dispersion relation: acoustic and optical branches

6.1 Dynamical matrix and branches

With the Bloch ansatz $u_{s,n} = (e_s/\sqrt{m_s})\mathrm{e}^{\mathrm{i}(kna-\omega t)}$ the $2N$ equations reduce, for each k, to $D(k)\boldsymbol{e} = \omega^2 \boldsymbol{e}$ with

$$D(k) = \begin{pmatrix} \dfrac{\sigma}{m_1} & -\dfrac{K_1 + K_2\mathrm{e}^{-\mathrm{i}ka}}{\sqrt{m_1 m_2}} \\ -\dfrac{K_1 + K_2\mathrm{e}^{\mathrm{i}ka}}{\sqrt{m_1 m_2}} & \dfrac{\sigma}{m_2} \end{pmatrix}, \qquad \sigma = K_1 + K_2. \tag{24}$$

It is Hermitian; its trace σ/μ does not depend on k and its determinant is $\det D = 4K_1K_2\sin^2(ka/2)/(m_1m_2)$, where we used $|K_1 + K_2\mathrm{e}^{\mathrm{i}ka}|^2 = \sigma^2 - 4K_1K_2\sin^2(ka/2)$. The eigenvalues are

$$\boxed{\omega_\pm^2(k) = \frac{\sigma}{2\mu}\left[1 \pm \sqrt{1 - \frac{16K_1K_2\,\mu^2}{\sigma^2\, m_1 m_2}\sin^2\frac{ka}{2}}\right]} \tag{25}$$

which for $K_1 = K_2 = K$ reduces to the familiar $\omega_\pm^2 = \frac{K}{\mu}\left[1 \pm \sqrt{1 - \frac{4\mu^2}{m_1m_2}\sin^2\frac{ka}{2}}\right]$.

In the code

`dispersion.omega_squared` evaluates the acoustic branch with Vieta's formula, $\omega_-^2 = \det D/\omega_+^2$, instead of subtracting two nearly equal numbers: with the subtraction, $\omega_-^2 \sim v_s^2k^2$ is lost to catastrophic cancellation for $ka \lesssim 10^{-8}$, precisely the region that dominates the thermodynamics at very low temperature.

6.2 Limits and gap

Long wavelengths ($k \to 0$). The acoustic branch is linear, $\omega_- \simeq v_s|k|$, with

$$v_s = a\sqrt{\frac{K_1K_2}{(K_1 + K_2)(m_1 + m_2)}}. \tag{26}$$

The interpretation is transparent: one cell of mass M joined to its neighbours by the two springs *in series*, of stiffness $K_1K_2/(K_1 + K_2)$. The optical branch tends to $\omega_+(0) = \sqrt{\sigma/\mu}$, the top of the spectrum.

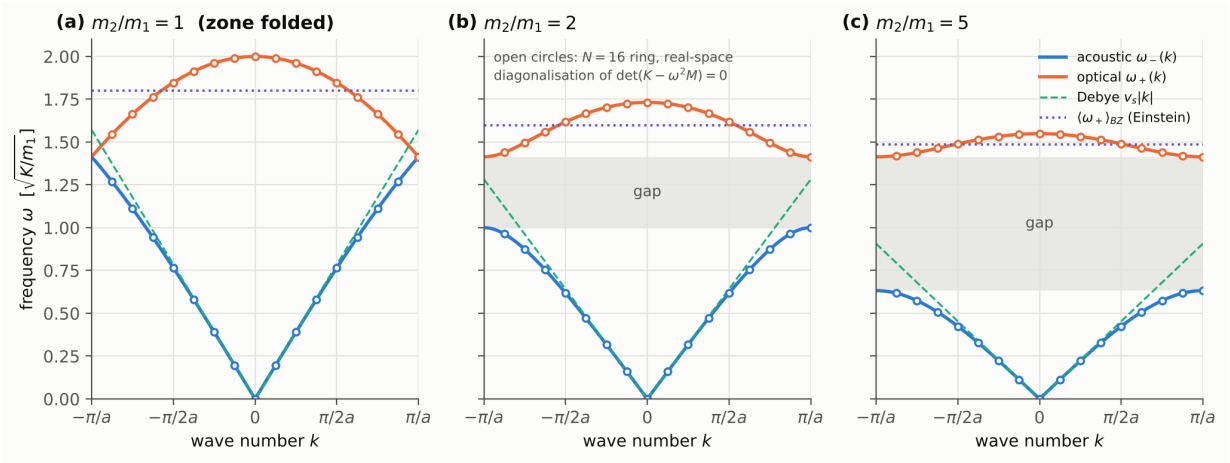


Figure 1: Acoustic (blue) and optical (orange) branches for three mass ratios with $K_1 = K_2$. Circles are the $2N$ frequencies of an $N = 16$ ring obtained by diagonalising $\det(K - \omega^2 M) = 0$ directly in real space, without Bloch's theorem: they fall on the curves at the N values of k allowed by Born–von Kármán (maximum difference $\sim 2 \times 10^{-8}$, round-off of the translation mode). Green: the Debye line $v_s|k|$; violet: the Einstein level $\langle\omega_+\rangle_{\mathrm{BZ}}$. (a) With equal masses the gap closes: this is the monatomic chain of spacing $a/2$ with its zone "folded".

Zone boundary ($k = \pi/a$). There $|D_{12}| = |K_1 - K_2|/\sqrt{m_1 m_2}$ and the two frequencies bound a **forbidden band** (gap). For $K_1 = K_2 = K$:

$$\omega_-(\pi/a) = \sqrt{\frac{2K}{m_{\max}}}, \qquad \omega_+(\pi/a) = \sqrt{\frac{2K}{m_{\min}}}, \tag{27}$$

and there are no normal modes between them. A perturbation driven at a gap frequency does not propagate: it decays exponentially.

For the reference chain ($m_1 = 1$, $m_2 = 2$, $K = 1$, $a = 1$): $v_s = 0.4082$, acoustic top 1, optical bottom $\sqrt{2} = 1.4142$, optical top $\sqrt{3} = 1.7321$ and gap 0.4142 (Fig. 1).

6.3 Eigenvectors: what moves in each branch

From the first row of $(D - \omega^2)\boldsymbol{e} = 0$, the ratio of physical amplitudes is

$$\frac{u_2}{u_1} = \frac{\sigma - m_1\omega^2}{K_1 + K_2 \mathrm{e}^{-\mathrm{i}ka}}. \tag{28}$$

- At $k = 0$, acoustic branch: $u_2/u_1 = 1$; the cell translates rigidly.
- At $k = 0$, optical branch: $u_2/u_1 = -m_1/m_2$; the atoms oscillate in antiphase and **the centre of mass of each cell stays at rest**.
- At $k = \pi/a$ (with $K_1 = K_2$, $m_2 > m_1$): in the acoustic branch only the heavy atoms move and in the optical branch only the light ones; each sublattice vibrates as if the other were a rigid wall, hence the frequencies $\sqrt{2K/m}$.

Fig. 2 shows the ratio (measured from each atom's own position, which for $K_1 = K_2$ makes it real: $\frac{2K - m_1\omega^2}{2K\cos(ka/2)}$) and the real-space patterns.

6.4 Limiting cases and generalisation

Equal masses, $m_1 = m_2$, $K_1 = K_2$. The gap closes and the two branches are the dispersion of the monatomic chain of spacing $a' = a/2$, $\omega = 2\sqrt{K/m}\,|\sin(qa'/2)|$, "folded" onto the smaller Brillouin zone of the doubled cell. Choosing a doubled cell is a decision of the observer[E5], not of the stage[E4]: the physics does not change.

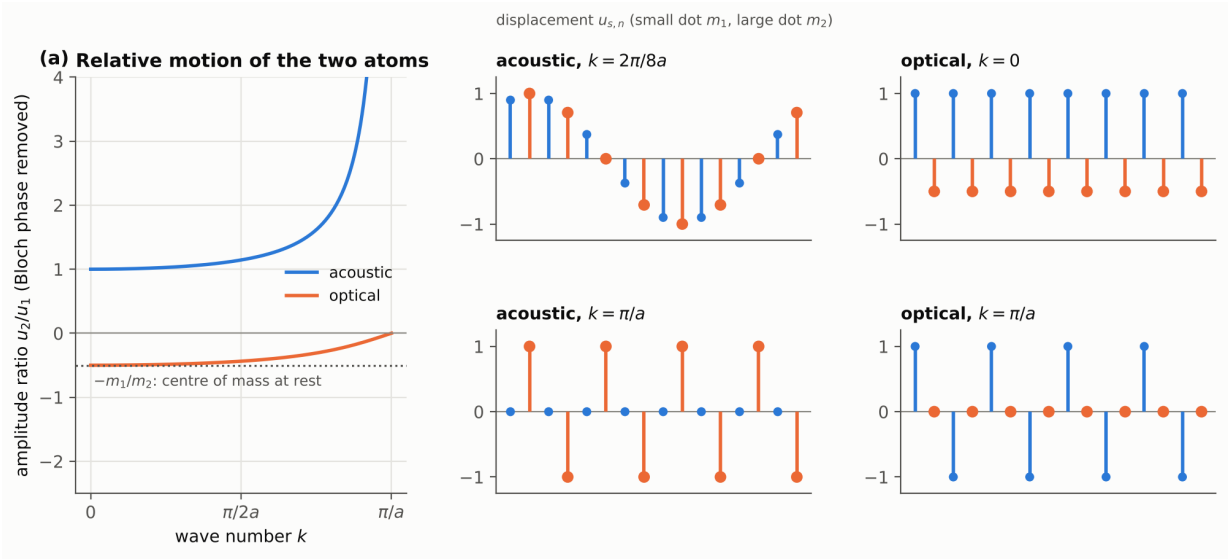


Figure 2: (a) Amplitude ratio u_2/u_1 across the Brillouin zone ($m_2/m_1 = 2$): positive (in phase) in the acoustic branch, negative (antiphase) in the optical branch, with the value $-m_1/m_2$ at $k = 0$. (b)–(e) Displacements of the 16 atoms of an 8-cell ring for four representative modes.

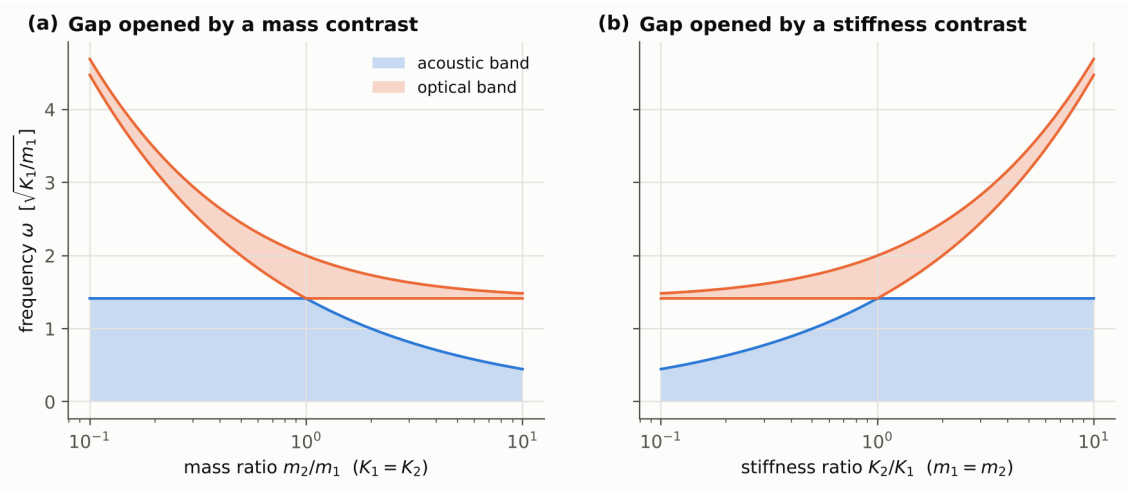


Figure 3: Allowed bands as a function of (a) the mass ratio with $K_1 = K_2$ and (b) the stiffness ratio with $m_1 = m_2$. In both cases the gap closes only when the two atoms of the cell are equivalent.

Large mass contrast, $m_2 \gg m_1$. The optical branch becomes almost flat, $\omega_+ \simeq \sqrt{2K/m_1}$: the light atoms vibrate almost independently in the "cage" of the heavy ones. This is exactly Einstein's hypothesis, which becomes almost exact for that branch. The acoustic branch tends to that of a monatomic chain of heavy atoms.

Stiffness contrast, $K_1 \neq K_2$. Even with equal masses a gap opens (Fig. 3b): it is enough that the true unit cell contains two non-equivalent atoms. This is the phonon version of a dimerised chain.

7 Density of states of the diatomic chain

7.1 Closed form

We invert the secular equation to express $s \equiv \sin^2(ka/2)$ as a function of ω. From $\det(D-\omega^2) = 0$, $(\sigma/m_1 - \omega^2)(\sigma/m_2 - \omega^2) = |D_{12}|^2$, and expanding the product the σ^2 term cancels exactly:

$$s(\omega) = \frac{m_1 m_2\, \omega^2 \left(\sigma/\mu - \omega^2\right)}{4K_1K_2}. \tag{29}$$

Since $ka = 2\arcsin\sqrt{s}$, $\mathrm{d}k/\mathrm{d}\omega = \frac{1}{a}\frac{s'(\omega)}{\sqrt{s(1-s)}}$ and, counting $\pm k$,

$$\boxed{g(\omega) = \frac{1}{\pi}\,\frac{|s'(\omega)|}{\sqrt{s(1-s)}}\,, \qquad s'(\omega) = \frac{m_1 m_2\, \omega\,(\sigma/\mu - 2\omega^2)}{2K_1K_2}\,, \qquad 0 < s(\omega) < 1} \tag{30}$$

and $g = 0$ outside the bands. Immediate checks:

- $\omega \to 0$: $g(0) = \frac{a}{\pi v_s}$, the flat DOS of an elastic medium (the one Debye uses).
- At the four band edges ($s = 1$, or $s' = 0$ at the optical top) g diverges as $(\omega_c - \omega)^{-1/2}$: van Hove singularities.
- The cumulative number of modes is also closed:

$$N(\omega) = \begin{cases} \frac{2}{\pi}\arcsin\sqrt{s(\omega)} & \text{acoustic band} \\ 1 & \text{gap} \\ 2 - \frac{2}{\pi}\arcsin\sqrt{s(\omega)} & \text{optical band} \end{cases} \tag{31}$$

  (the optical branch *decreases* with $|k|$), with $N(\infty) = 2$: one acoustic and one optical mode per cell.

In the code

The DOS is computed along three independent routes in `dos.py`: the closed form (`analytic_dos`), the histogram of the frequencies on a fine k-grid (`sampled_dos`) and the histogram of the $2N$ eigenvalues of a finite ring diagonalised in real space (`histogram_dos`). Since $N(\omega)$ is exact, the exact average of g over each histogram bin is $\Delta N/\Delta\omega$ (`bin_averaged_dos`): the maximum difference with the sampling of 2×10^5 k values is 3×10^{-4}, and all three routes are normalised to 2 up to round-off.

7.2 Physical reading of Fig. 4

The exact DOS has three features no simple model reproduces at once: the acoustic plateau at low frequency (which Debye does capture), the pile-up of modes at band edges (van Hove) and the gap. Debye places modes where there are none (between 1.73 and 2.57) to make up the total count; Einstein places all modes at a single frequency. Since C_V is an average of g weighted by $c(\hbar\omega/k_\mathrm{B}T)$, a smooth function, the fine details of g are washed out: the quantitative question is *by how much*.

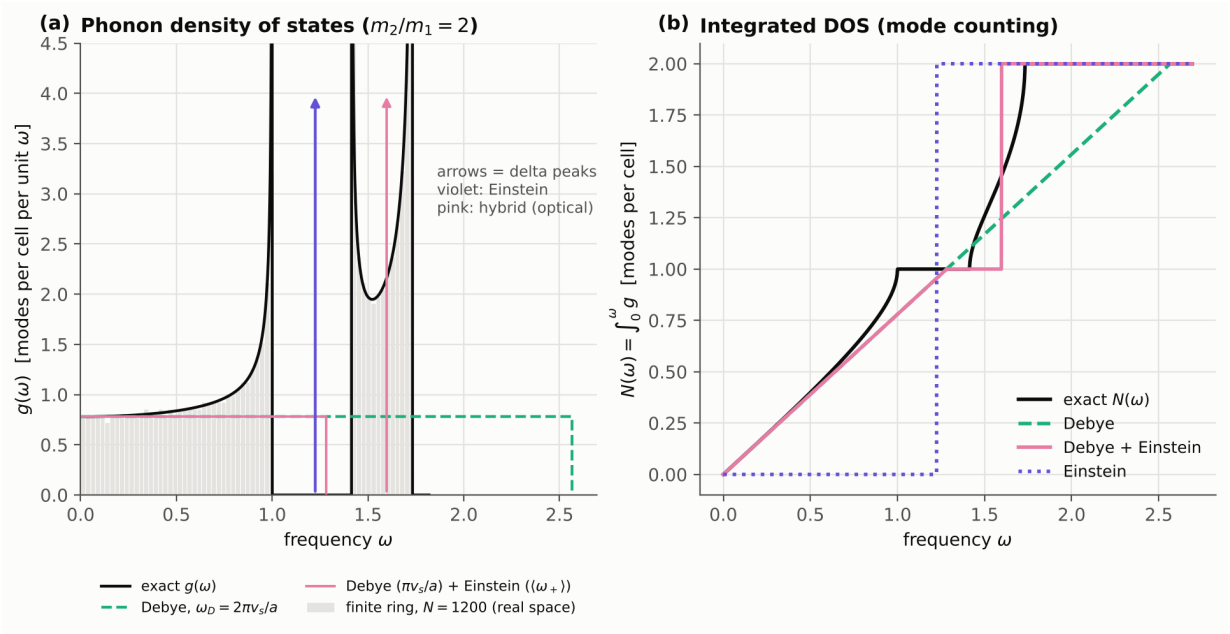


Figure 4: (a) Exact DOS (black) versus the histogram of a 1200-cell ring (grey), Debye's flat DOS (green), the hybrid model (pink: Debye for the acoustic branch up to $\pi v_s/a$ plus an Einstein peak at $\langle\omega_+\rangle$) and the Einstein peak (violet). (b) Cumulative number of modes: Debye spreads the two modes per cell uniformly up to $2\pi v_s/a = 2.57$, well above the real top 1.73.

8 Heat capacity: numerical result versus the models

8.1 Numerical method

Instead of integrating (12) over ω, where g is singular, we integrate over k, where everything is smooth:

$$\frac{C_V}{Nk_\mathrm{B}} = \frac{a}{\pi}\int_0^{\pi/a} \mathrm{d}k \sum_{b=\pm} c\left(\frac{\hbar\omega_b(k)}{k_\mathrm{B}T}\right), \tag{32}$$

with Gauss–Legendre quadrature (exponential convergence). At low temperature the acoustic integrand is only appreciable for $k \lesssim 40k_\mathrm{B}T/\hbar v_s$, so the interval is split there to concentrate nodes where the function lives. The result agrees with the integral over the analytic DOS (`scipy.integrate.quad`) to 10^{-12} and with the exact sum over the modes of a 400-cell ring to 10^{-4} (finite-size effect).

8.2 Results

$k_\mathrm{B}T$	exact	Debye	Einstein	Debye + Einstein
0.01	0.02566	0.02565	2×10^{-49}	0.02565
0.1	0.2629	0.2565	0.00144	0.2565
0.3	0.8553	0.7654	0.5816	0.8141
0.5	1.3286	1.1931	1.2411	1.3023
1.0	1.7764	1.6952	1.7677	1.7684
2.0	1.9393	1.9129	1.9387	1.9372

Table 1: C_V/Nk_B of the reference chain. The models use parameters fixed by the physics (not fitted): $\Theta_D = 2\pi\hbar v_s/ak_\mathrm{B} = 2.565$, $\Theta_E = \sqrt{\langle\omega^2\rangle} = 1.225$ and, in the hybrid, $\omega_D^{\mathrm{ac}} = \pi v_s/a = 1.283$ and $\omega_E = \langle\omega_+\rangle_{\mathrm{BZ}} = 1.597$.

The quantitative conclusions for $m_2/m_1 = 2$ are:

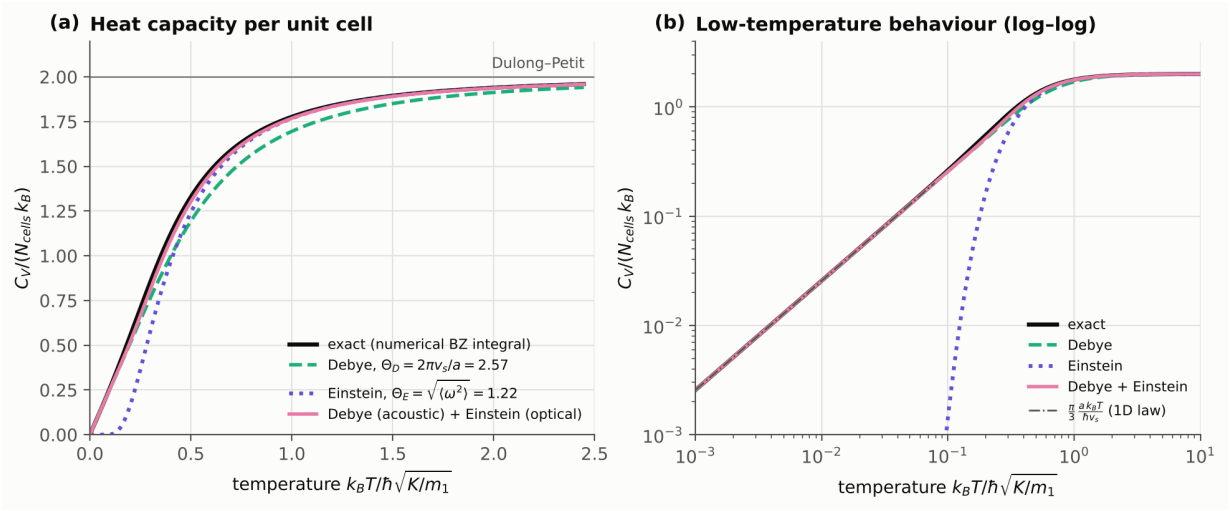


Figure 5: Heat capacity per cell of the reference chain ($m_2/m_1 = 2$). (a) Linear scale: exact (black) versus Debye with $\Theta_D = 2\pi v_s/a$, Einstein with $\Theta_E = \sqrt{\langle\omega^2\rangle}$ and the hybrid. (b) Log–log scale: at low T the exact result follows the linear law (16) (numerical slope 2.56510 versus the theoretical $\pi a/3v_s = 2.56510$) while Einstein drops exponentially.

1. **Low temperature.** Debye (and hence the hybrid) is exact; the relative error is 2×10^{-4} at $k_BT = 10^{-2}$ and 2×10^{-6} at $k_BT = 10^{-3}$. Einstein fails completely (49 orders of magnitude at $k_BT = 0.01$).
2. **High temperature.** All three tend to Dulong–Petit. Einstein with $\omega_E^2 = \langle\omega^2\rangle$ reproduces by construction the $1/T^2$ correction (at $k_BT = 10$: exact 1.997503, expansion 1.997500) and is the best of the three above $k_BT \approx 1$.
3. **Intermediate region** ($k_BT \sim 0.1$–0.5, where C_V rises fastest). Here lives the physics of the real DOS: Debye underestimates C_V by up to 11% because it spreads modes up to 2.57, where they are still frozen; the hybrid reduces the maximum error to 6%.
4. **Fits.** Least-squares fitting a single Θ over $k_BT \in [0.02, 3]$: Debye gives $\Theta_D = 2.256$ with rms error 0.020, but then it is no longer exact at low T (maximum relative error 14%); Einstein gives $\Theta_E = 0.962$ with rms error 0.14 and still fails qualitatively at low T. A good global fit does not imply good physics: the fitted parameter hides the structure of $g(\omega)$.

8.3 Branch decomposition

Since the system is ideal, $C_V = C_{ac} + C_{op}$ exactly (Fig. 6). The optical branch is frozen below $k_BT \approx 0.1$ (it contributes 0.02% of C_V at $k_BT = 0.1$), 10% at $k_BT = 0.25$ and 34% at $k_BT = 0.5$. All low-temperature physics is acoustic, which justifies the hybrid model: Debye where the dispersion is linear, Einstein where the branch is narrow. The figure also shows where the hybrid's error lives: in the acoustic part, because the real branch bends (and piles up modes at its top) while Debye keeps it linear.

8.4 Dependence on the mass ratio

Fig. 7 summarises how the quality of the models changes with m_2/m_1:

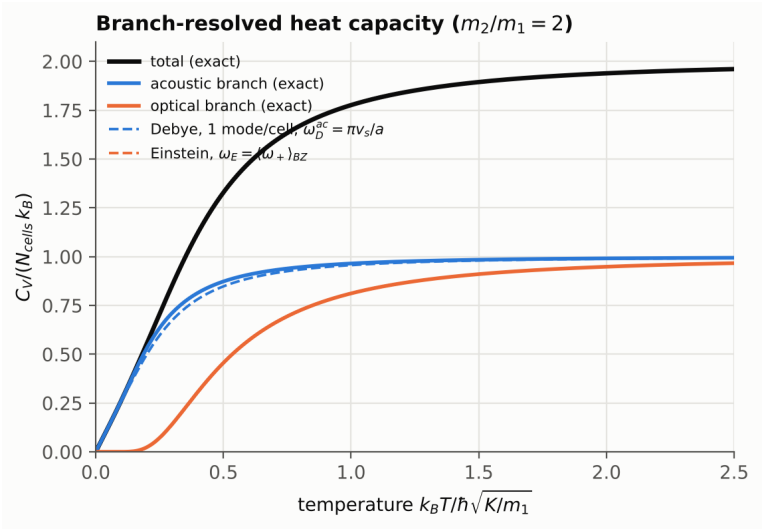


Figure 6: Acoustic and optical contributions to the heat capacity (each saturates at k_B per cell) and the Debye (1 mode/cell) and Einstein models applied to each branch separately.

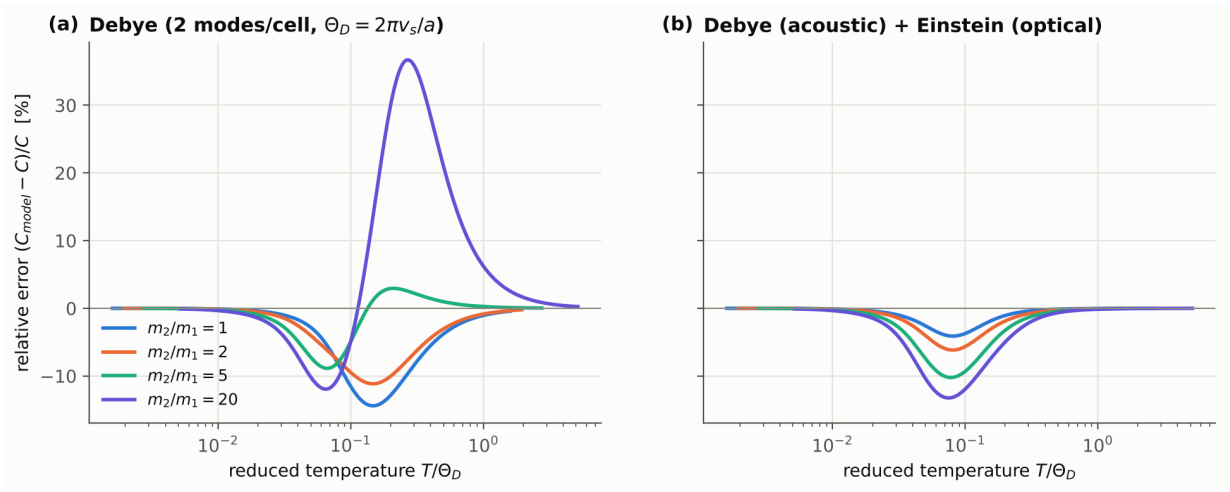


Figure 7: Relative error of (a) Debye and (b) the hybrid model versus T/Θ_D for several mass ratios.

m_2/m_1	max. error Debye	max. error Debye + Einstein
1	14.4%	4.1%
2	11.2%	6.2%
5	8.9%	10.2%
20	36.7%	13.2%

- **Debye** degrades dramatically for large contrasts: the heavier the second atom, the smaller v_s, the smaller Θ_D and the worse it describes the optical branch, whose frequency $\sqrt{2K/m_1}$ does not depend on m_2. For $m_2/m_1 = 20$ it overestimates C_V by 37% (it lets optical modes vibrate that are actually frozen).
- **The hybrid** shows an apparently paradoxical trend: its error *grows* with m_2/m_1 even though the optical branch gets flatter and flatter (Einstein becomes more and more exact). The reason is that the residual error is Debye's error on the acoustic branch, which for large contrast is that of a monatomic chain of heavy atoms, sinusoidal and with a strong van Hove singularity at its top; its Debye error is $\simeq 14\%$, the same as for the monatomic chain in the first row. For $m_2/m_1 = 1$ the hybrid is very good because the "optical branch" is the folded upper half of the same sinusoid and $\langle\omega_+\rangle$ represents it well.

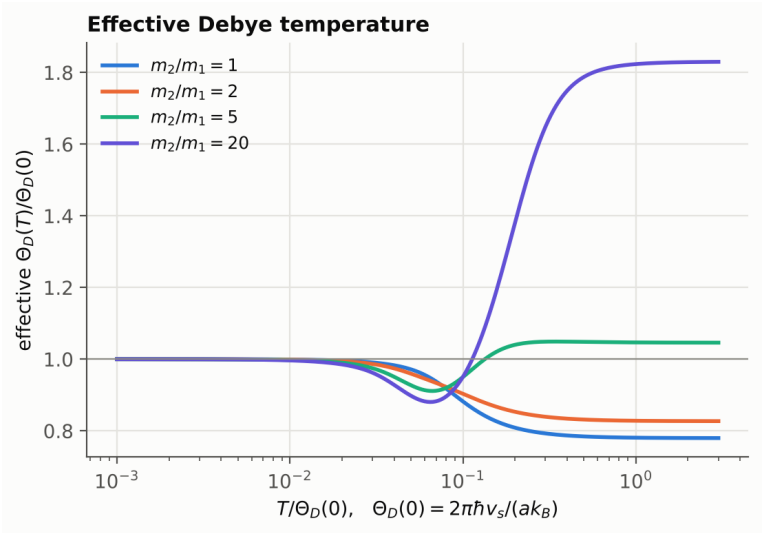


Figure 8: Effective Debye temperature $\Theta_D(T)$, normalised to its $T = 0$ value, for several mass ratios.

8.5 Effective Debye temperature

Fig. 8 shows $\Theta_D(T)/\Theta_D(0)$. All curves start at 1 (Debye exact as $T \to 0$) and drop as soon as the acoustic modes near the top of the band are excited: the real DOS piles them up below the Debye cutoff (van Hove) and the crystal "softens". For $m_2/m_1 = 1$ and 2 the drop is monotonic; for $m_2/m_1 = 5$ and 20 there is a minimum near $T \approx 0.06\,\Theta_D(0)$ followed by a rise, when the optical branch unfreezes. At high temperature all of them saturate at $\sqrt{3\,\langle\omega^2\rangle}/\Theta_D(0)$: 0.78, 0.83, 1.05 and 1.83 for $m_2/m_1 = 1, 2, 5, 20$. For large contrasts the saturation exceeds 1: the optical branch, high in frequency and holding half of the modes, "stiffens" the crystal compared with what the sound velocity predicts. This is exactly what is observed experimentally in solids with very different masses in the basis, and the reason why a single tabulated Debye temperature never describes the whole temperature range well.

9 Other observables

9.1 Internal energy, entropy and free energy

Fig. 9 shows U, S and F per cell in the quantum and classical treatments. Two statistical-mechanics remarks: (i) the negative classical entropy at low T is the same symptom as the negative Sackur–Tetrode entropy at low temperature: discretising phase space into cells of volume h stops making sense when the typical action is of order $\hbar$; (ii) thermodynamic consistency $S = -\partial F/\partial T$, $C_V = \partial U/\partial T$, $U = F + TS$ is checked numerically in the tests to 10^{-6}.

9.2 Displacement fluctuations: no long-range order in 1D

This is the most interesting observable physically. Expanding in normal modes, $u_{s,n} = N^{-1/2}\sum_{k,b}(e_s/\sqrt{m_s})\mathrm{e}^{\mathrm{i}kna}Q_{k,b}$, and using that each normal coordinate is a unit-mass oscillator with

$$\langle|Q_{k,b}|^2\rangle = \frac{\hbar}{2\omega}\coth\frac{\hbar\omega}{2k_{\mathrm{B}}T} \xrightarrow{\hbar\to 0} \frac{k_{\mathrm{B}}T}{\omega^2}, \tag{33}$$

we compute the **mean-square relative displacement** of two atoms r cells apart,

$$G_{ss'}(r) = \langle(u_{s',n+r} - u_{s,n})^2\rangle = \frac{a}{2\pi}\int_{\mathrm{BZ}}\mathrm{d}k\sum_b \langle|Q_{k,b}|^2\rangle\left|\frac{e_{s'}\mathrm{e}^{\mathrm{i}kra}}{\sqrt{m_{s'}}} - \frac{e_s}{\sqrt{m_s}}\right|^2. \tag{34}$$

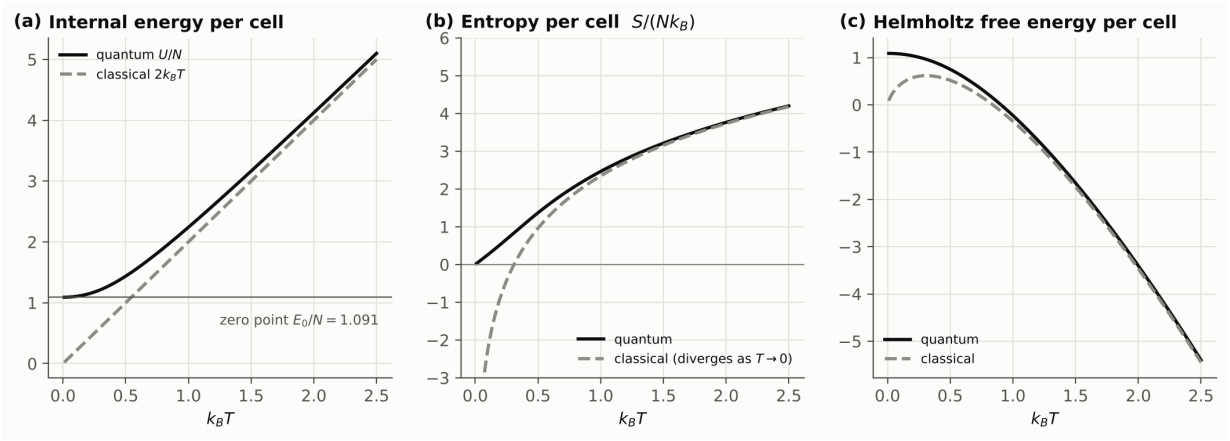


Figure 9: (a) Internal energy per cell: at $T = 0$ the zero-point energy $E_0/N = 1.091$ remains; at high T the quantum curve tends to the classical $2k_{\mathrm{B}}T$ (not to $2k_{\mathrm{B}}T + E_0/N$: the zero point cancels in the expansion $\frac{\hbar\omega}{2}\coth\frac{y}{2} = k_{\mathrm{B}}T + \frac{\hbar^2\omega^2}{12k_{\mathrm{B}}T} + \dots$). (b) Entropy: the quantum one vanishes at $T = 0$ (third law); the classical one diverges to $-\infty$, a sign that classical counting of microstates fails at low temperature. (c) Helmholtz free energy.

The factor $|\cdots|^2 \propto k^2$ of the acoustic branch as $k \to 0$ makes $G(r)$ finite for each r, but its dependence on r is very revealing (Fig. 10a):

- **Classically** the result is exact and simple: $G_{11}(r) = r\,k_{\mathrm{B}}T\left(\frac{1}{K_1} + \frac{1}{K_2}\right)$. The bond stretches are independent Gaussian variables of variance $k_{\mathrm{B}}T/K_i$ (equipartition applied to each spring), and the separation between two atoms is the sum of $2r$ of them: a random walk. The code reproduces it with relative error $< 10^{-8}$.
- **Quantum, at** $T = 0$, $G(r) \simeq \frac{\hbar a}{\pi M v_s}\ln r$: logarithmic growth (numerical slope 0.259899 versus the theoretical 0.259899).

Since $G(r) \to \infty$ as $r \to \infty$, the positions of distant atoms are uncorrelated: **a 1D chain has no long-range positional order** at any temperature (Landau–Peierls argument). It is the same moral as the absence of a phase transition in the 1D Ising model: in one dimension long-wavelength fluctuations always win. Equivalently, the mean-square displacement of one atom in a ring of N cells (with the rigid translation removed) diverges with N (Fig. 10b): $\langle u_1^2\rangle = \frac{N}{12}k_{\mathrm{B}}T\left(\frac{1}{K_1} + \frac{1}{K_2}\right)$ classically (checked: $0.166667\,Nk_{\mathrm{B}}T$ for our parameters) and $\simeq \frac{\hbar a}{2\pi M v_s}\ln N$ at $T = 0$. The Debye–Waller factor $\mathrm{e}^{-q^2\langle u^2\rangle}$ vanishes in the thermodynamic limit: a 1D crystal would not give sharp Bragg peaks.

9.3 Bond fluctuation

Local fluctuations, on the other hand, are well defined: the fluctuation of the K_1 bond, $\langle(u_{2,n} - u_{1,n})^2\rangle$, is finite at every temperature (Fig. 10c). At $T = 0$ it equals 0.545 (in units of $\hbar/\sqrt{Km_1}$) because of zero-point motion, and at high T it tends to equipartition, $k_{\mathrm{B}}T/K_1$. This is the quantity entering a Lindemann-type melting criterion, and it again shows the crossover from the quantum regime (plateau) to the classical one (straight line).

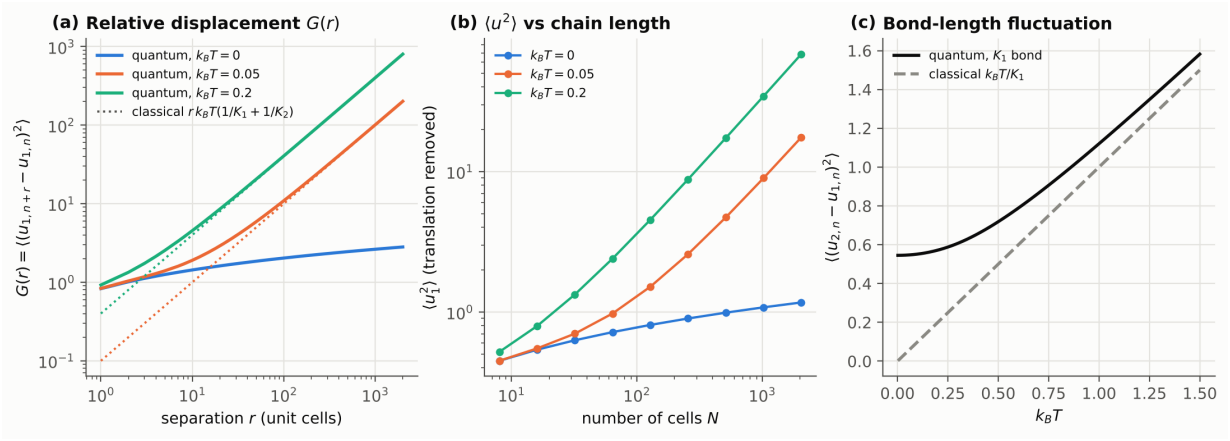


Figure 10: (a) Quantum relative displacement $G(r)$ at three temperatures (solid) and the classical result $rk_{\mathrm{B}}T(1/K_1 + 1/K_2)$ (dotted): at short distances the zero point dominates, at long distances the thermal linear growth. (b) $\langle u_1^2 \rangle$ in rings of N cells: linear in N at $T > 0$, logarithmic at $T = 0$. (c) Bond fluctuation: the zero point (0.545 at $T = 0$) versus the classical line $k_{\mathrm{B}}T/K_1$.

10 Independent validation: Langevin dynamics

10.1 Motivation

Everything above uses normal modes. To test the physics along a completely different route we simulate the *classical* dynamics of a ring of atoms coupled to a heat bath through the Langevin equation:

$$\mathrm{d}q_i = \frac{p_i}{m_i}\mathrm{d}t, \qquad \mathrm{d}p_i = F_i(q)\,\mathrm{d}t - \gamma p_i\,\mathrm{d}t + \sqrt{2m_i\gamma k_{\mathrm{B}}T}\;\mathrm{d}W_i, \tag{35}$$

with $\mathrm{d}W_i$ Wiener increments (white noise). The dissipative force $-\gamma p$ and the stochastic one balance on average (Einstein relation between friction and diffusion) and the stationary distribution is the canonical one.

As stressed in §2.2, an isolated harmonic chain is integrable: a single microcanonical (NVE) trajectory would never sample the canonical ensemble because each mode keeps its energy. The Langevin bath supplies the "residual interaction" that thermalises each mode.

10.2 Integrator

We use the BAOAB splitting (Leimkuhler and Matthews, 2013):

$$\text{B: } p \mathrel{+}= \tfrac{\Delta t}{2}F, \quad \text{A: } q \mathrel{+}= \tfrac{\Delta t}{2}\tfrac{p}{m}, \quad \text{O: } p \leftarrow c\,p + \sqrt{(1-c^2)mk_{\mathrm{B}}T}\,\xi, \quad \text{A}, \quad \text{B}, \qquad c = \mathrm{e}^{-\gamma\Delta t},$$

with independent normal ξ. For $\gamma = 0$ it reduces exactly to the **velocity Verlet** algorithm, which is symplectic (it preserves phase-space volume, consistently with Liouville's equation); the test `test_velocity_verlet_conserves_energy` checks energy conservation to 10^{-3} after 5000 steps. With $\gamma > 0$, BAOAB samples configurational averages with a very small time-step error.

10.3 Three measurements and their exact predictions

DOS from the velocity spectrum. By the Wiener–Khinchin theorem, the spectral density of each atom's velocity is the Fourier transform of its autocorrelation. In equilibrium every mode has mean kinetic energy $k_{\mathrm{B}}T/2$ (equipartition), so the mass-weighted spectrum $\sum_i m_i|\tilde{v}_i(\omega)|^2$ is proportional to $g(\omega)$, broadened by friction (each line is approximately a Lorentzian of half-width $\gamma/2$). With $N = 256$ cells (mode spacing $\approx 0.01 < \gamma = 0.03$) the spectrum reproduces the broadened exact DOS, including the van Hove singularities and the gap (Fig. 11a).

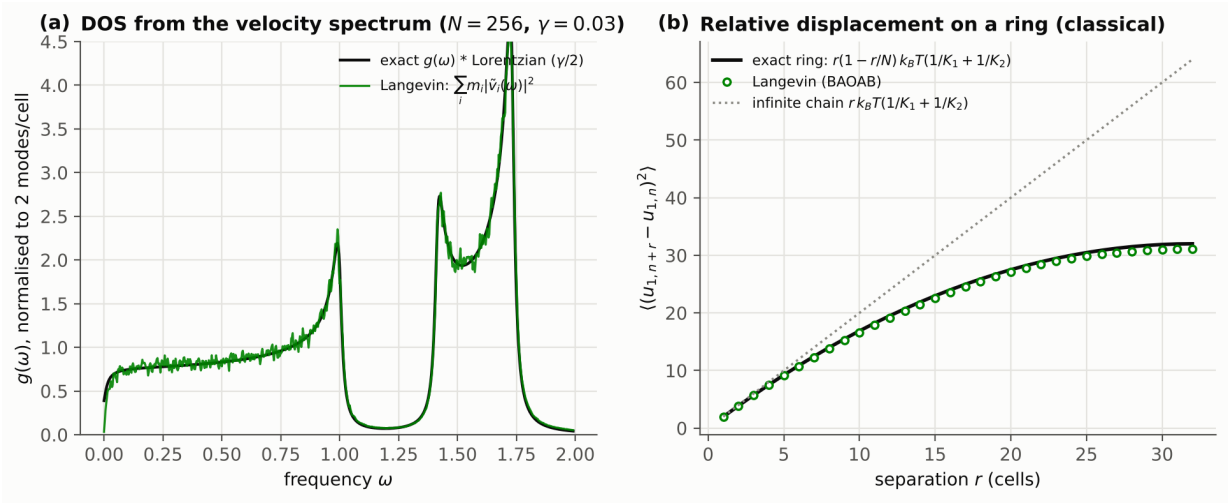


Figure 11: Langevin dynamics (BAOAB, $\Delta t = 0.05$, $k_{\mathrm{B}}T = 1$). (a) Mass-weighted velocity spectrum ($N = 256$, $\gamma = 0.03$) versus the exact DOS convolved with a Lorentzian of half-width $\gamma/2$. (b) Relative displacement on a 64-cell ring ($\gamma = 0.5$) versus the exact Brownian-bridge prediction and the infinite-chain straight line.

Heat capacity from fluctuations. Through the fluctuation–response relation, $C_V = (\Delta E)^2/k_{\mathrm{B}}T^2$. For a classical ring of N cells, counting quadratic terms ($2N$ kinetic and $2N - 1$ independent potential ones, because the stretches sum to zero around the ring), the exact value is $C_V/Nk_{\mathrm{B}} = 2 - 1/2N = 1.9922$ for $N = 64$. The simulation gives 1.956 ± 0.023 (block-averaging error).

Relative displacement on the ring. On a ring the stretches are independent Gaussians conditioned to sum to zero: a discrete "Brownian bridge", with

$$G(r) = r\left(1 - \frac{r}{N}\right) k_{\mathrm{B}}T \left(\frac{1}{K_1} + \frac{1}{K_2}\right), \tag{36}$$

which the simulation reproduces with a maximum relative deviation of 3% (Fig. 11b). The kinetic temperatures of the two sublattices are 0.998 and 0.996 (equipartition, independent of mass) and the bond variances 0.991 and 0.989 versus the exact 0.992.

11 Numerical implementation

11.1 Code structure

Module	Content
`parameters.py`	`ChainParameters`: immutable `dataclass` with m_1, m_2, K_1, K_2, a and derived quantities (v_s, gap, $\langle\omega^2\rangle$, …).
`dispersion.py`	$D(k)$, closed-form branches (Vieta form), eigenvectors, amplitude ratio, group velocity, Born–von Kármán grid.
`real_space.py`	M and K matrices of a ring or open chain and solution of $\det(K - \omega^2 M) = 0$ (`scipy.linalg.eigh`).
`dos.py`	Closed-form DOS, closed-form $N(\omega)$, histograms (k-grid and real space), broadened DOS, Debye DOS.
`thermodynamics.py`	Numerically stable single-oscillator functions (`expm1`), Gauss–Legendre quadrature in k, U, C_V, F, S, branch decomposition, asymptotics.
`models.py`	1D Debye function, Debye, Einstein and hybrid models, characteristic frequencies, fits, $\Theta_D(T)$.
`observables.py`	$G_{ss'}(r)$ with composite quadrature, $\langle u_s^2\rangle$ on finite rings, bond fluctuations.
`langevin.py`	BAOAB integrator, full simulation with estimators and exact ring predictions.
`plotting.py`	Common figure style.

11.2 Numerical subtleties

- **Cancellations.** ω_-^2 is computed as $\det D/\omega_+^2$ and $s(\omega)$ in its expanded form; both avoid subtracting nearly equal numbers and keep machine relative precision as $k \to 0$ or $\omega \to 0$. The eigenvectors from `eigh` are combined with the closed-form eigenvalues for the same reason.
- **Integrating over k, not ω.** The integrand in k is analytic; in ω it has $(\omega_c - \omega)^{-1/2}$ singularities.
- **Low temperature.** The Gauss–Legendre grid is split at $k_c = 40 k_\mathrm{B} T/\hbar v_s$. The slope C_V/T at $k_\mathrm{B}T = 10^{-4}$ matches the theoretical one to 7 digits.
- **Oscillating integrands.** $G(r)$ contains $\mathrm{e}^{\mathrm{i}kra}$; a composite rule is used with a number of panels growing with r (linear cost in r, instead of the cubic cost of a high-order Gauss–Legendre rule).
- **Reproducibility.** Langevin noise uses `numpy.random.Generator` with a fixed seed; every number in this document comes from `scripts/make_figures.py`, which stores it in `results/results.json`.

11.3 Tests

`tests/test_physics.py` contains 26 tests comparing the code with *independent* results: closed form versus diagonalisation, real space versus Bloch, acoustic sum rule, monatomic limit, trace and second moment, normalisation and derivative of $N(\omega)$, limits of C_V, DOS versus k-quadrature versus finite ring, thermodynamic consistency, limits of the Debye function, $\Theta_D(\infty)$, exact classical random walk, logarithmic growth at $T = 0$, equipartition and energy conservation in Verlet. Run them with `pytest` or with `python tests/run_tests.py`.

12 Discussion and conclusions

1. **From the general to the particular.** The discrete translation symmetry of the stage[E4] turns a problem of $2N$ coupled degrees of freedom into N problems of size 2×2; the har-

monic approximation turns the crystal into an ideal system of oscillators, and the canonical ensemble makes the whole thermodynamics depend only on $g(\omega)$. The diatomic chain is the minimal example in which all the ingredients of a real crystal appear: acoustic and optical branches, gap, van Hove singularities and a DOS that is neither Debye's nor Einstein's.

2. **Branches.** The acoustic branch is the dynamical manifestation of translational invariance (sum rule); the optical branch exists because there is more than one atom per cell, and the gap because those atoms are not equivalent (different mass or stiffness).

3. **Heat capacity.** Debye is exact at low temperature (only v_s matters) and Einstein with $\omega_E^2 = \langle\omega^2\rangle$ is exact in the first high-temperature correction. Neither describes the intermediate region, which is where the real DOS shows up: maximum errors of 11% (Debye) and 6% (hybrid) for $m_2/m_1 = 2$.

4. **Models and mass ratio.** Debye degrades quickly with mass contrast (37% for $m_2/m_1 = 20$) because the optical scale is not set by the sound velocity. The hybrid greatly improves that case but keeps an irreducible $\sim 14\%$ error, Debye's error on a sinusoidal acoustic branch. The effective Debye temperature varies between 0.78 and 1.83 times its $T = 0$ value depending on the contrast: a single Θ_D does not characterise a solid with a basis.

5. **Fluctuations.** In 1D there is no long-range positional order: $G(r) \propto r$ at $T > 0$ and $\propto \ln r$ at $T = 0$, in analogy with the absence of a phase transition in the 1D Ising model. Local (bond) fluctuations are finite and show the crossover from the quantum regime (zero point) to the classical one (equipartition).

6. **Validation.** A Langevin simulation, which does not use normal modes, reproduces the DOS (velocity spectrum), Dulong–Petit (through energy fluctuations) and the displacement correlations. It also illustrates that an isolated harmonic crystal is not ergodic: equilibrium requires the bath (or anharmonicity).

Natural extensions. Chains with $p > 2$ atoms per cell, second-neighbour interactions, defects and impurities (localised modes inside the gap, detectable with the already implemented inverse participation ratio), anharmonicity (thermal expansion, thermal conductivity, the FPUT problem) and the step to 2D and 3D, where the Debye DOS is no longer flat and the T^3 law appears.

A Derivation of the sound velocity

Expanding $\omega_-^2 = \det D/\omega_+^2$ for $k \to 0$: $\det D \simeq K_1K_2a^2k^2/(m_1m_2)$ and $\omega_+^2 \simeq \sigma/\mu$, so

$$\omega_-^2 \simeq \frac{K_1K_2\,\mu}{\sigma\, m_1m_2}\, a^2k^2 = \frac{K_1K_2}{(K_1+K_2)(m_1+m_2)}\, a^2k^2.$$

B Second moment of the DOS

Since $\operatorname{tr} D(k) = \sigma/m_1 + \sigma/m_2 = \sigma/\mu$ for every k,

$$\langle\omega^2\rangle = \frac{1}{2N}\sum_k \left(\omega_-^2 + \omega_+^2\right) = \frac{1}{2N}\sum_k \operatorname{tr} D(k) = \frac{\sigma}{2\mu}.$$

For the reference chain, $\langle\omega^2\rangle = 1.5$ and $\sqrt{\langle\omega^2\rangle} = 1.2247$.

C Classical relative displacement

In the infinite chain the Hamiltonian written in terms of the momenta and the stretches ϵ_j of each bond is a sum of independent quadratic terms, so $\left\langle \epsilon_j^2 \right\rangle = k_{\mathrm{B}}T/K_j$ and $\langle \epsilon_j \epsilon_l \rangle = 0$ for $j \neq l$. The difference $u_{1,n+r} - u_{1,n}$ is the sum of r bonds K_1 and r bonds K_2, hence $G_{11}(r) = r k_{\mathrm{B}} T (1/K_1 + 1/K_2)$. On a ring, the constraint $\sum_j \epsilon_j = 0$ turns the walk into a Brownian bridge and the factor $(1 - r/N)$ appears.

D Reproducibility

```
git clone https://github.com/jesus-garcia-izquierdo/diatomic-chain
cd diatomic-chain
pip install -r requirements.txt
python tests/run_tests.py          # 26 tests
python scripts/make_figures.py     # figures + results/results.json
```

References

[1] P. García González, *Termodinámica y Física Estadística II*, lecture notes, BSc in Physics, Universidad Autónoma de Madrid (2024–2025).

[2] *Mecánica Estadística*, MSc lecture notes (chapters 1–4), Universidad Autónoma de Madrid.

[3] *Classical and quantum mechanics: the five elements of a physical problem*, course notes.

[4] N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, Saunders (1976), chs. 22–23.

[5] C. Kittel, *Introduction to Solid State Physics*, 8th ed., Wiley (2005), chs. 4–5.

[6] R. K. Pathria and P. D. Beale, *Statistical Mechanics*, 3rd ed., Elsevier (2011), ch. 7.

[7] B. Leimkuhler and C. Matthews, "Rational construction of stochastic numerical methods for molecular sampling", *Appl. Math. Res. Express* **2013**, 34–56.

[8] R. Peierls, *Quantum Theory of Solids*, Oxford (1955).